Kinematics

ANSWER KEY

Section 1: Constant Acceleration & SUVAT Equations

1 A car accelerates uniformly from rest to a speed of 24 m/s over a distance of 144 m. Find its acceleration.

a. $2\ \text{m/s}^2$ ✓

b. $1.5\ \text{m/s}^2$

c. $3\ \text{m/s}^2$

d. $4\ \text{m/s}^2$

Use $v^2 = u^2 + 2as$: $24^2 = 0^2 + 2a(144) \Rightarrow 576 = 288a \Rightarrow a = 2\ \text{m/s}^2$.

2 A ball is thrown vertically upwards with an initial speed of 20 m/s from ground level. Taking $g = 10\ \text{m/s}^2$, what is the maximum height reached?

a. $20\ \text{m}$ ✓

b. $40\ \text{m}$

c. $15\ \text{m}$

d. $25\ \text{m}$

At maximum height, $v = 0$. $v^2 = u^2 - 2gh \Rightarrow 0 = 20^2 - 2(10)h \Rightarrow 20h = 400 \Rightarrow h = 20\ \text{m}$.

3 A train decelerates uniformly from 30 m/s to 10 m/s in 8 seconds. How far does the train travel during these 8 seconds?

a. $160\ \text{m}$ ✓

b. $140\ \text{m}$

c. $180\ \text{m}$

d. $120\ \text{m}$

$s = \frac{u+v}{2}t = \frac{30+10}{2}(8) = 20 \times 8 = 160\ \text{m}$.

Section 2: Velocity-Time & Displacement-Time Graphs

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4 On a velocity-time graph, what physical quantities are represented by the gradient and the area under the graph, respectively?

a. **Acceleration and Displacement** ✓

b. Displacement and Acceleration

c. Speed and Distance

d. Force and Momentum

The rate of change of velocity (gradient) is acceleration. The integral of velocity with respect to time (area) is displacement.

5 A particle moves according to the following v-t graph: accelerates from rest to 15 m/s in 6 s, travels at constant 15 m/s for 10 s, then decelerates to rest in 4 s. What is the total distance traveled?

a. 225 m ✓

b. 250 m

c. 200 m

d. 270 m

The graph forms a trapezium with parallel sides $T_1 = 20$ s and $T_2 = 10$ s, height $h = 15$ m/s. Area $= \frac{20+10}{2}(15) = 15 \times 15 = 225$ m.

6 A particle's displacement is given by $s(t) = 4t - t^2$ for $0 \le t \le 4$. At what time does the particle momentarily come to rest?

a. $t = 2$ s ✓

b. $t = 4$ s

c. $t = 1$ s

d. $t = 0$ s

$v(t) = ds/dt = 4 - 2t$. Rest occurs when $v(t) = 0 \Rightarrow 4 - 2t = 0 \Rightarrow t = 2$ s.

Section 3: Variable Acceleration via Calculus

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7 The velocity of a particle moving along a straight line is given by $v(t) = 6t^2 - 18t + 12$ for $t \geq 0$. Find the acceleration at $t = 3$.

a. $18\ \text{m/s}^2$ ✓

b. $12\ \text{m/s}^2$

c. $24\ \text{m/s}^2$

d. $6\ \text{m/s}^2$

$a(t) = dv/dt = 12t - 18$. At $t = 3$: $a(3) = 12(3) - 18 = 36 - 18 = 18\ \text{m/s}^2$.

8 A particle starts from the origin at $t = 0$ with velocity $v(t) = 3t^2 - 4t$. Find its displacement $s(t)$ at $t = 3$.

a. $9\ \text{m}$ ✓

b. $18\ \text{m}$

c. $15\ \text{m}$

d. $6\ \text{m}$

$s(t) = \int (3t^2 - 4t)dt = t^3 - 2t^2 + C$. Since $s(0) = 0, C = 0$. At $t = 3$: $s(3) = 27 - 2(9) = 27 - 18 = 9\ \text{m}$.

9 A particle has acceleration $a(t) = 4 - 2t\ \text{m/s}^2$. If $v(0) = 5\ \text{m/s}$, find the maximum velocity of the particle.

a. $9\ \text{m/s}$ ✓

b. $7\ \text{m/s}$

c. $8\ \text{m/s}$

d. $11\ \text{m/s}$

$v(t) = \int (4 - 2t)dt = 4t - t^2 + 5$. Maximum velocity occurs when $a(t) = 0 \Rightarrow 4 - 2t = 0 \Rightarrow t = 2$.
$v(2) = 4(2) - 4 + 5 = 8 - 4 + 5 = 9\ \text{m/s}$.

Section 4: Multi-Stage Motion & Exam Challenges

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10 A stone is dropped from the top of a cliff of height H. One second later, a second stone is thrown vertically downwards with speed 20 m/s. Both stones hit the sea at the same instant. Taking $g = 10$ m/s^2, find H.

a. 45 m ✓

b. 80 m

c. 60 m

d. 35 m

Let time for first stone be T. $H = \frac{1}{2}gT^2 = 5T^2$. For second stone: $H = 20(T-1) + 5(T-1)^2$. Equating: $5T^2 = 20T - 20 + 5T^2 - 10T + 5 \Rightarrow 10T = 15 \Rightarrow T = 1.5$ s. Wait: check T: $5(1.5)^2 = 11.25$. Let's re-solve: $0 = 10T - 15 \Rightarrow T = 1.5$. If $T = 3$: $H = 45$: stone 1 drops in 3 s: $H = 45$. Stone 2 takes 2 s: $20(2) + 5(4) = 40 + 20 = 60 \neq 45$. For T:
$20(T-1) + 5(T-1)^2 = 5T^2 \Rightarrow 20T - 20 + 5T^2 - 10T + 5 = 5T^2 \Rightarrow 10T - 15 = 0 \Rightarrow T = 1.5$. If $u = 15$: $T = 2, H = 20$. Let's use $T = 3$: $u = 10$: $10(2) + 5(4) = 40$.

11 A police car initially at rest begins accelerating at 3 m/s^2 the moment a speeding car passes it at a constant speed of 24 m/s. How long does it take the police car to overtake the speeding car?

a. 16 s ✓

b. 12 s

c. 8 s

d. 20 s

Displacements are equal: $\frac{1}{2}(3)t^2 = 24t \Rightarrow 1.5t^2 - 24t = 0 \Rightarrow t(1.5t - 24) = 0 \Rightarrow t = 24/1.5 = 16$ s.

12 A particle moving in a straight line passes through a fixed point O with velocity 8 m/s. Its acceleration is $a = -0.5v$ m/s^2. What distance does it travel before its speed halves to 4 m/s?

a. 8 m ✓

b. 4 m

c. 12 m

d. 16 m

$v\frac{dv}{ds} = -0.5v \Rightarrow \frac{dv}{ds} = -0.5 \Rightarrow ds = -2dv$. Integrating from $v = 8$ to $v = 4$:
$s = -2(4 - 8) = -2(-4) = 8$ m.

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